

Spiderbot PCB Design Considerations

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1 DCDC converter

1.1 Input capacitor

I choose a ceramic input capacitor with the capacity $10\mu F$. I found the value recommended in the datasheet and I did not bother to calculate it. Hope it's good.

1.2 Minimum duty cycle

This is needed for the inductor calculation.

$$D_{min} = \frac{V_{OUT} + V_F}{V_{INMAX} - V_{SW}} = \frac{3.3 + 0.6}{12.6 - 0.6} = 0.31 \quad (1)$$

Where V_F is the forward voltage of the diode and V_{SW} is the voltage drop of the MOSFET inside the buck converter (I guess $0.3A \cdot 200m\Omega$)

1.3 Inductor

I am choosing to have 20% current ripple from the maximum current of 3A that the converter can supply. That is 0.6A.

Then the minimum inductance is:

$$L_{MIN} = \frac{V_{OUT} + V_F}{\Delta I_{MAX}} \cdot \frac{1 - D_{MIN}}{F_{SW}} = \frac{3.3 + 0.6}{0.6} \cdot \frac{1 - 0.31}{500000} = 9\mu H \quad (2)$$